Q03 Discussion Questions

1. If a sufficiently good SFT dataset existed (perfect demonstrations for every prompt type), would RLHF still add anything? What specifically do preferences buy that demonstrations cannot?
2. The KL anchor is described as a trust region for the reward model rather than a safety feature. Construct a concrete scenario where a large β (tight anchor) *causes* a worse outcome than a small one.
3. DPO's derivation eliminates the reward model by making it implicit. What do we lose operationally when we can no longer inspect or reuse r_ϕ — and could that loss matter for safety evaluation?
4. GRPO's group-relative advantage assumes the group contains useful signal (some wins, some losses). What happens to training when a task is either trivially solvable or unsolvable for the current policy — and what would you do about it?
5. "Preference optimization aligns to raters, not truth." Whose preferences were collected for the models you use, and what systematic biases would you expect to survive into deployment because of it?